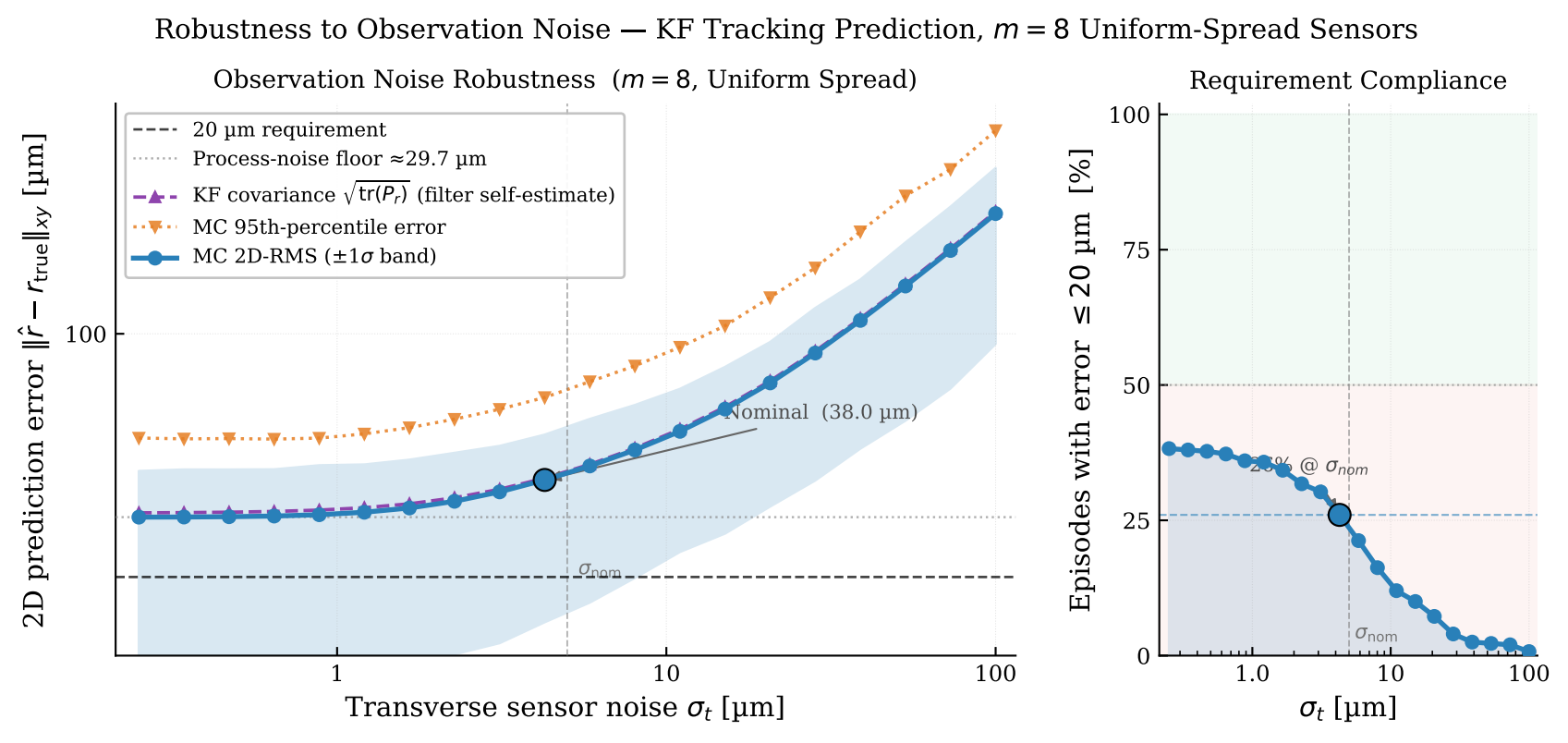
Robustness to Observation Noise — KF Tracking Prediction, $m = 8$ Uniform-Spread Sensors
Observation Noise Robustness ($m = 8$, Uniform Spread)
20 µm requirement
Process-noise floor ≈29.7 µm
KF covariance $\sqrt{\mathrm{tr}(P_r)}$ (filter self-estimate)
MC 95th-percentile error
MC 2D-RMS (±1σ band)
Nominal (38.0 µm)
σ_{nom}
2D prediction error $\|\hat{r} - r_{true}\|_{xy}$ [µm]
100
1
10
100
Transverse sensor noise σ_t [µm]
Requirement Compliance
Episodes with error ≤ 20 µm [%]
100
75
50
25
0
@ σ_{nom}
σ_{nom}
1.0
10
100
σ_t [µm]